

Error at Different Reading Accuracies

emateer03

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As the uncertainty in the dipstick reading increases, the resulting volume error increases proportionally. Figure (1) compares the impact of different reading uncertainties on the percentage error in volume, demonstrating how measurement precision directly affects reliability.

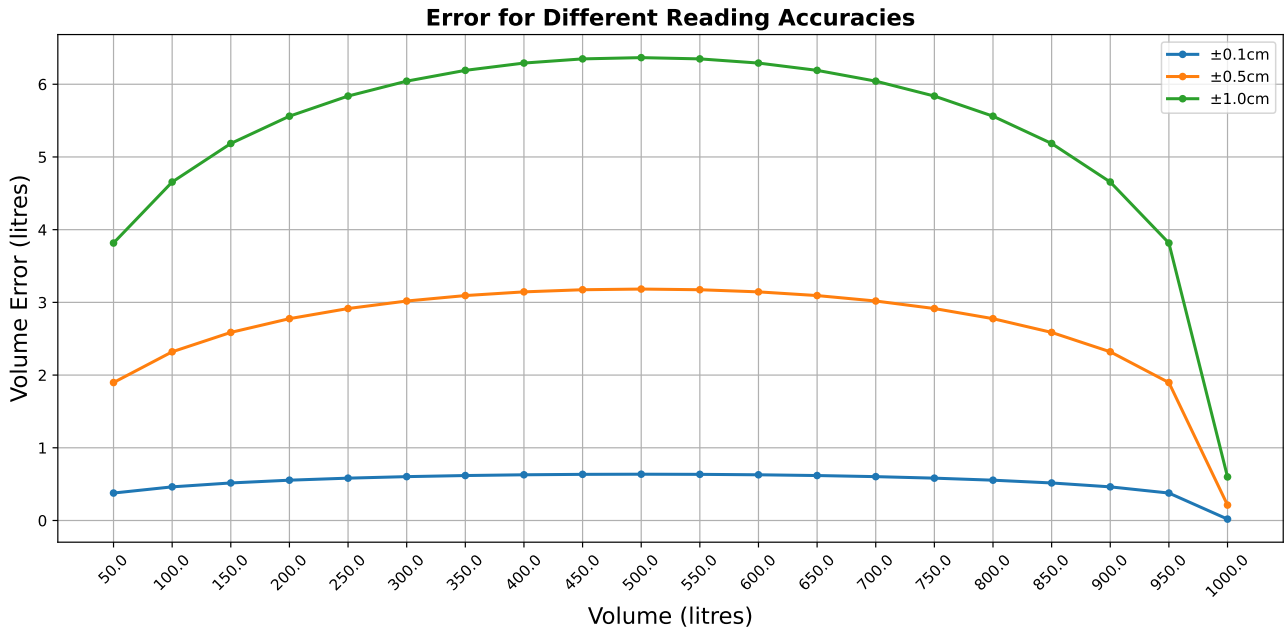


Figure 1: Absolute volume error versus fill level for three levels of dipstick reading uncertainty $\pm 0.1\text{cm}$ (blue), $\pm 0.5\text{cm}$ (orange) and $\pm 1.0\text{cm}$ (green).

(see [0] for the code for the graph above)

Elliemat21 (2025). AMA3020SoloProject/DipSticksProjectCode.ipynb at main · Elliemat21/AMA3020SoloProject. [online] GitHub.https://github.com/Elliemat21/AMA3020_SoloProject/blob/main/DipSticks_Project_Code.ipynb (accessed 25/02/26)